

Teaching Statement

I have taught mathematics across a wide range of levels and to students with very different mathematical backgrounds, from introductory Calculus and proof-based Linear Algebra to Analysis and Optimization. I adjust the pace, structure, assignments, and assessments of my courses based on what my students need. Students have often commented on my enthusiasm in the classroom and my willingness to act on their feedback.

Throughout these adjustments, my main goal remains the same. I want to make serious mathematical thinking accessible without removing its intellectual demands. I want students to build confidence in fundamental skills while moving beyond computation toward conceptual understanding, critical judgment, and rigorous reasoning.

Teaching Students with Different Mathematical Backgrounds

Students enter mathematics courses with different levels of computational skill, experience with proofs, and comfort with abstract ideas. Their goals also vary, from preparing for advanced mathematics to completing their final mathematics requirement. I therefore give students different ways to approach the material. I expect every student to understand the main ideas, while students with stronger preparation have opportunities to explore the material in greater depth through advanced questions.

These differences are especially clear in Analysis and Optimization, which enrolls students from various STEM fields. Although linear algebra is a prerequisite, students enter the course with very different levels of preparation, and many have never written a formal proof. When students had difficulty with proofs, I began breaking arguments into smaller steps during class. I also redesigned some homework assignments as guided worksheets that gave students the structure of an argument but asked them to supply definitions, justifications, and intermediate steps. Students later said that proofs felt more accessible and intuitive and that these changes helped them understand the material.

I use students' questions and written work to judge what the class is ready to do independently. When many students have the same difficulty, I review the topic or provide more structure before asking them to work independently again. This allows me to support students when they need it without lowering the expectations of the course.

Organizing Mathematics Around Ideas Rather Than Procedures

I organize my courses around a few central mathematical ideas rather than presenting the material as a sequence of procedures. This helps students connect individual topics and see how concrete problems lead to more abstract ideas. Before introducing a theorem or method, I ask what problem motivates it, why the result should be plausible, and what makes it nontrivial. I return to these questions throughout the course. They help students see the course as one connected story rather than a collection of separate topics. They also help students recognize when familiar methods can be used in new settings.

In Analysis and Optimization, approximation is the main organizing idea. I return throughout the semester to the principle that linear approximation helps locate possible extrema, while quadratic approximation classifies local behavior. Students then see the gradient, Hessian, and Newton's method as tools for answering related questions about local models rather than as separate topics. I use concrete optimization problems to introduce the theory and show why its hypotheses matter and why more precise local models are needed.

In Linear Algebra, I organize the course around questions such as why matrix multiplication has its particular definition, how the solvability of $A\mathbf{x} = \mathbf{b}$ changes with $\mathbf{b}$, and how abstract linear transformations can be represented by matrices. These questions lead naturally to abstract concepts. Students encounter these concepts as answers to concrete problems rather than as definitions introduced without motivation. They also learn to ask whether an object exists, whether it is unique, and how it can be classified or represented.

In this way, students learn when and why standard procedures work instead of treating them as recipes to memorize.

Designing Assessment as Part of Learning

I design assessments not only to measure learning but also to shape how students study. Homework provides low-stakes practice, short quizzes encourage regular preparation, and exams measure students' independent understanding and reasoning. In my current Analysis and Optimization course, quiz questions are selected from assigned homework problems, and quiz performance can count toward the computational part of the corresponding exam. This gives students credit for regular practice with core computational skills and allows me to focus more of each exam on concepts and proofs.

I distinguish among routine computation, conceptual understanding, and proof writing. Core questions assess essential concepts and computations, while more theory-based problems ask students to explain their reasoning clearly. Multiple assessments give students opportunities to recover from an early setback and show improvement over time. When I use oral examinations, I give students sample problems and possible follow-up questions in advance. I do this so that the exam measures their preparation, reasoning, and communication rather than how they respond to an unexpected question.

I continue to revise my assessments based on student performance. In earlier courses, I used exam corrections to encourage students to revisit their mistakes. More recently, I have used more frequent supervised assessments and closer connections among homework, quizzes, and exams to give students earlier and more consistent feedback.

Teaching Mathematics in a Changing Environment

The growing use of AI has made it more important to distinguish between producing a mathematical answer and understanding it. When I discuss AI with students, I emphasize that learning mathematics is essential to using AI properly and responsibly. This requires more than writing effective prompts. Students must be able to examine assumptions, find gaps in an argument, and decide whether a conclusion follows. Learning mathematics develops these skills because students must justify each step, follow the logic of an argument, and check the results for themselves. These skills become even more important as AI becomes more common.

I also consider this when I design homework and assessments. I generally require students to submit homework by hand. Even when students have used external tools, writing out a solution gives them another opportunity to work through and verify the reasoning before submitting it. I treat homework mainly as practice and use supervised assessments to evaluate what students understand on their own. Technology may change how students produce answers, but my goal is to teach them how to decide whether those answers can be trusted.

Across my courses, I want students to master fundamental skills and use them to ask productive questions, write clear arguments, and judge mathematical claims for themselves. I adjust my teaching to help students with different starting points reach these goals. These experiences have prepared me to teach a broad range of undergraduate mathematics courses.